

Android P&Q Environment Building Guide

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Revision History

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This is the initial release.

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1. Android Environment

This document builds an Android compilation environment based on the Ubuntu 18.04 Desktop environment. The required environment configuration is as follows.

- Environment: Ubuntu 18.04 Desktop
- CPU: X86_64, 4 cores
- Memory requirement: 16G
- Hard disk requirements: at least 250G

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2. Environment Setup Step

2.1 Basement Environment and Tools

```
sudo dpkg --add-architecture i386
sudo apt-get update
sudo apt-get install -y automake make perl gcc g++
sudo apt-get install -y bison git gperf libxml2-utils zip unixodbc dos2unix java-common vim autofs
sudo apt-get install -y repo python-pip python-dev
```

2.2 32/64 bit System Compatible Tool & Uboot Tools

```
sudo apt-get install -y libc6:i386 zlib1g:i386 lib32stdc++6 libstdc++6:i386
sudo apt-get install -y u-boot-tools
```

2.3 Installing Openjdk

```
sudo add-apt-repository ppa:openjdk-r/ppa
sudo apt-get install -y openjdk-8-jdk
sudo update-alternatives --config java
sudo update-alternatives --config javac
```

2.4 Toolchain Configuration

You can get toolchain from FAE. Different platforms use different toolchains, but they are configured in the same way:

1. Install toolchain in /opt/ directory.
2. Set variable PATH to access toolchain binary. You can set following command in /etc/bashrc or ~/.bashrc.

```
export PATH=/opt/<Toolchain Rootdir>/bin:$PATH
```

3. Run <toolchain>-gcc -v, and it indicates ok when print version.

The following lists the used toolchains:

	Module	Toolchain	
		Toolchain name	Version
Android P	BL33	aarch64-none-elf-gcc	4.8.3
	Kernel	64bit aarch64-linux-gnu-gcc	6.3.1
		32bit arm-linux-gnueabi-gcc	6.3.1
Android Q	BL33	aarch64-none-elf-gcc	4.9.2
		arm-none-eabi-gcc	4.4.1
	Kernel	64bit aarch64-linux-gnu-gcc	6.3.1
		32bit arm-linux-gnueabi-gcc	6.3.1

3. Q&A

1. Build fail with: failed with: signal: killed

It might be the swap partition is too small, and following set 1GB swap area:

```
dd if=/dev/zero of=/root/swapfile bs=1M count=1024
mkswap /root/swapfile
swapon /root/swapfile
```

If you want to set swap area when system start, add following statement in /etc/fstab:

```
/root/swapfile swap swap defaults 0 0
```

2. Build fail with: temporary area no space

Compressing system.new.dat with brotli

```
running: brotli --quality=6 --output=/tmp/tmpi0WKJ3/system.new.dat.br
/tmp/tmpi0WKJ3/system.new.dat
```

Traceback (most recent call last):

File "build/make/tools/releasetools/ota_from_target_files", line 2004, in <module> main(sys.argv[1:])

File "build/make/tools/releasetools/ota_from_target_files", line 1978, in main output_file=args[1]

File "build/make/tools/releasetools/ota_from_target_files", line 832, in WriteFullOTApackage system_diff.WriteScript(script, output_zip)

File "/home/amlogic/android/android-p/build/make/tools/releasetools/common.py", line 1580, in WriteScriptself._WriteUpdate(script, output_zip)

File "/home/amlogic/android/android-p/build/make/tools/releasetools/common.py", line 1735, in _WriteUpdate 'compression of {}.new.dat failed'.format(self.partition)

AssertionError: compression of system.new.dat

Building Android needs /tmp/ (variable \$TMPDIR) directory to save temporary files, so it will report error when temporary directory has no space. For this problem, you can just redirect \$TMPDIR to others folder with enough space:

```
mkdir -p /home/<homedir>/tmp
export TMPDIR=/home/<homedir>/tmp
```

If you want to auto setup when system start, please set this commands in ~/.bashrc or /etc/bashrc.

3. Exception in thread "main" java.lang.OutOfMemoryError: Java heap space

Java JVM memory is not enough.

```
export JAVA_OPTS='-Xms1024m -Xmx1024m'
```

-Xms <size> Set JVM heap size when initialize.

-Xmx <size> Set JVM heap max memory size.

If you want to auto setup when system start, please set this command in ~/.bashrc or /etc/bashrc.